

MILESTONES

DEVELOPS INTEREST

Moves to England in 1757; becomes interested in astronomy and observes the night sky.

FINDS NEW PLANET

In 1781, discovers Uranus and is elected a Fellow of the Royal Society; now a professional astronomer.

OBSERVES NEBULAE

Publishes first of three papers on nonstellar objects in 1786; redefines understanding of nebulae.

BUILDS TELESCOPE

In 1789, constructs an enormous, 40-ft (12-m) telescope with a 47-in (122-cm) mirror.

INFRARED RAYS

Discovers, in 1800, that the Sun emits invisible "calorific rays," now known as infrared rays.

A German-born astronomer living in England, William Herschel discovered Uranus, the first new planet to be found since the invention of the telescope. He was a skillful engineer who designed and built the world's largest telescope at the time.

William Herschel's interest in astronomy began after he learned how to make telescopes in order to study the night sky. Determined to look beyond the closest planets, he built a powerful telescope through which he could study the farthest celestial bodies. Herschel made detailed observations of the night sky, methodically dividing it into quarters and recording his sightings systematically. In 1781, he noted a small disk, which was confirmed as the seventh planet in the solar system. Named Uranus, its discovery led to the eventual sighting of Neptune in 1846.

Further discoveries

Astronomy in the 19th century profited from Herschel's considerable contributions. From his observations, he argued that nebulae (clouds of gas and dust) are composed of stars, not fluid, and he went on to catalog 2,500 nebulae and star clusters and 848 double stars. He discovered the two moons of Uranus, as well as two new moons of Saturn, and was the first to discover infrared rays after making studies of the Sun. Herschel's exceptional skill as a telescope builder, combined with his meticulous observations of the night sky, made him one of the greatest visual astronomers of all time.

"I have **looked further** into **space** than ever human being **did before** me."

William Herschel, 1781

Herschel captured the public imagination with his telescope, built in 1789, but it worked poorly and he considered it a failure.

